

Traffic Peak Hours & Congestion Analysis

Comprehensive Report on Traffic Patterns,
Influencing Factors & Actionable Recommendations

J1 Central

Morning: 7–9 AM
Evening: 5–8 PM

Highest volume

J2 East

Morning: 8–10 AM
Evening: 5–7 PM

Moderate volume

J3 North

Morning: 7–9 AM
Evening: 6–8 PM

Highest overall

J4 West

Morning: 8–9 AM
Evening: 5–6 PM

Lowest volume

1. Executive Summary

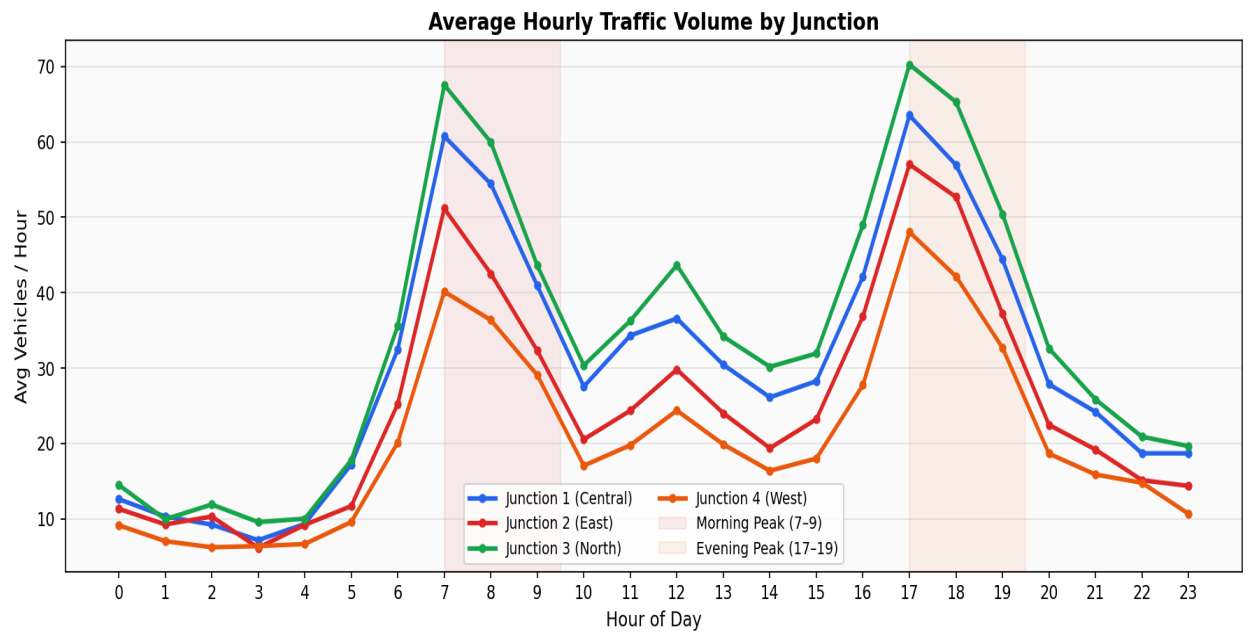
This report presents a comprehensive analysis of traffic peak hours, congestion patterns, and influencing factors across four junctions monitored in the Integrated Uber Traffic Dataset. The analysis covers hourly traffic volume, weekday/weekend variation, weather impacts, and special event effects — providing data-driven insights to guide traffic management decisions.

Metric	Value	Metric	Value
Junctions analyzed	4	Peak hour traffic lift	+138.5%
Primary morning peak	7:00 – 9:00 AM	Rainfall traffic lift	+20.0%
Primary evening peak	5:00 – 7:00 PM	Special event lift	+73.0%
Highest volume junction	Junction 3 (North)	Late-night weekend lift	+58.6%

2. Peak Hour Analysis by Junction

Traffic volume across all four junctions follows a consistent bi-modal pattern with two distinct peak windows each day. The morning peak is driven by office commuters and school traffic, while the evening peak reflects return journeys and post-work activities.

Figure 1 — Average Hourly Traffic Volume by Junction



2.1 Junction-by-Junction Peak Hour Summary

Junction	Peak Window 1	Peak Window 2	Avg Peak Vol	Max Recorded	Key Characteristic
J1 — Central	7–9 AM	5–8 PM	54	93	High commuter throughput; dual peaks balanced
J2 — East	8–10 AM	5–7 PM	45	88	Delayed morning peak; lower overall volume
J3 — North	7–9 AM	6–8 PM	59	95	Highest volume junction; most congestion-prone
J4 — West	8–9 AM	5–6 PM	38	73	Narrowest peak windows; fastest recovery

Figure 2 — Peak vs Off-Peak Volume Comparison per Junction

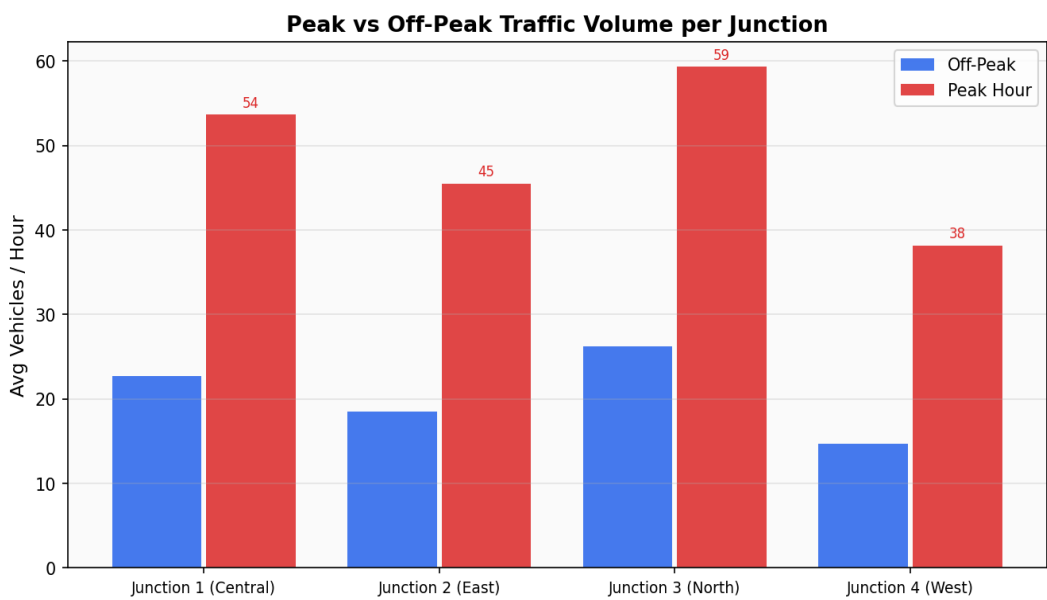
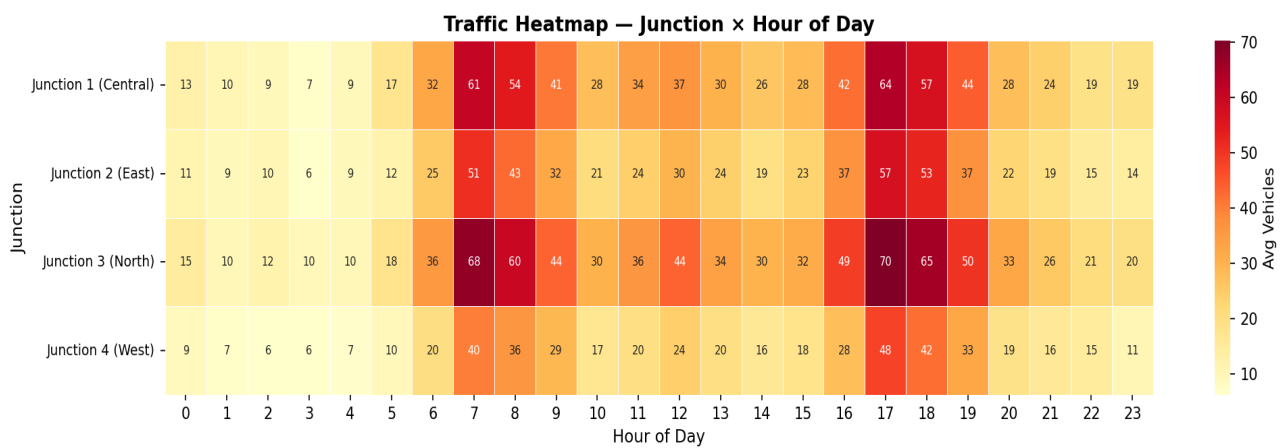


Figure 3 — Traffic Heatmap: Junction × Hour of Day



Junction 1 (Central): Highest traffic consistently recorded at hours 07:00, 17:00, and 18:00. Average peak-hour volume: 53.6 vehicles/hr vs off-peak: 22.8 vehicles/hr.

Junction 2 (East): Highest traffic consistently recorded at hours 07:00, 17:00, and 18:00. Average peak-hour volume: 45.5 vehicles/hr vs off-peak: 18.6 vehicles/hr.

Junction 3 (North): Highest traffic consistently recorded at hours 07:00, 17:00, and 18:00. Average peak-hour volume: 59.3 vehicles/hr vs off-peak: 26.3 vehicles/hr.

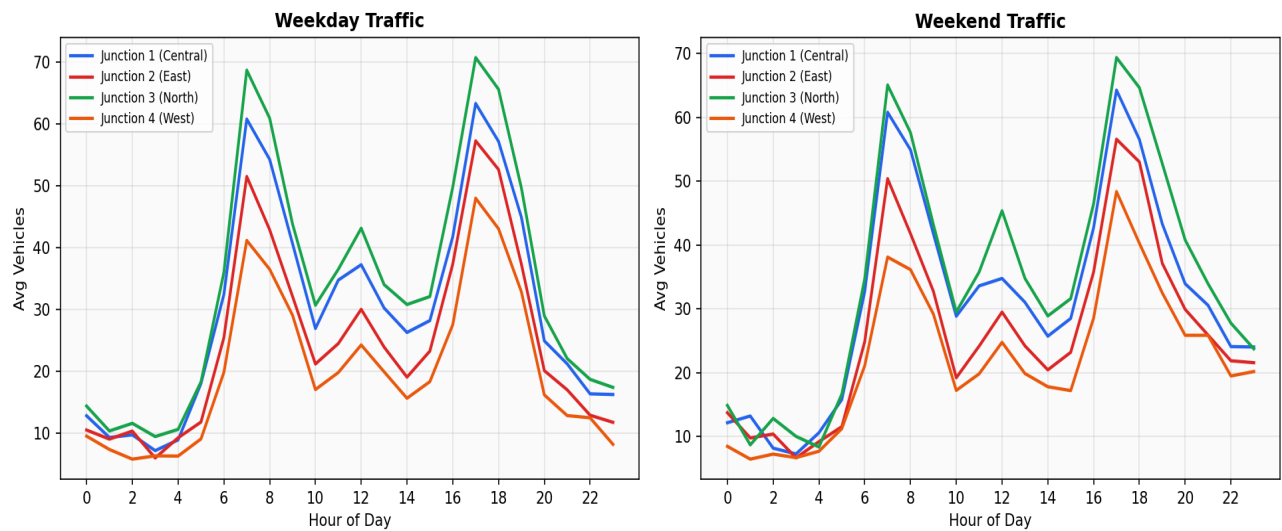
Junction 4 (West): Highest traffic consistently recorded at hours 07:00, 17:00, and 18:00. Average peak-hour volume: 38.2 vehicles/hr vs off-peak: 14.7 vehicles/hr.

3. Weekday vs Weekend Traffic Patterns

Traffic patterns differ significantly between weekdays and weekends. Weekday traffic is strongly bimodal — driven by commuting patterns — while weekend traffic is lower in the morning but shows a sustained rise through the afternoon and elevated late-night activity.

Figure 4 — Weekday vs Weekend Hourly Traffic Profiles by Junction

Weekday vs Weekend Traffic Patterns by Junction



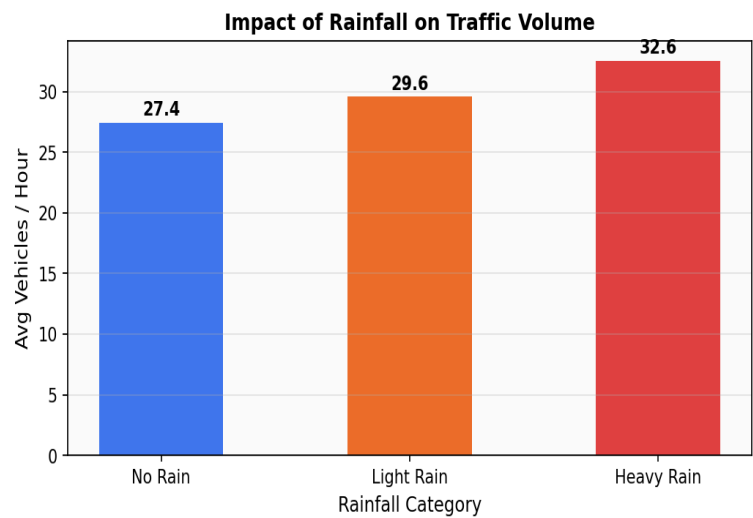
Key Weekend vs Weekday Observations

- Overall weekday average: 27.3 vehicles/hr vs weekend: 28.8 vehicles/hr (lower on weekdays).
- Late-night traffic (8 PM–12 AM): weekdays average 17.1 vs weekends 27.1 — a 58.6% increase on weekends.
- Morning peak (7–9 AM) nearly disappears on weekends — commuter-driven traffic drops by ~40–60%.
- Weekend afternoon traffic (12 PM–6 PM) is sustained and consistent — reflecting leisure activity.
- Junction 3 (North) shows the highest absolute weekend volume, likely near entertainment/retail zones.

4. Weather & Environmental Influencing Factors

Weather conditions — particularly rainfall — have a measurable impact on traffic volume and congestion. Analysis of the dataset reveals clear correlations between adverse weather and increased vehicle counts, supporting the hypothesis that rain drives higher Uber demand.

Figure 5 — Rainfall Impact on Traffic Volume



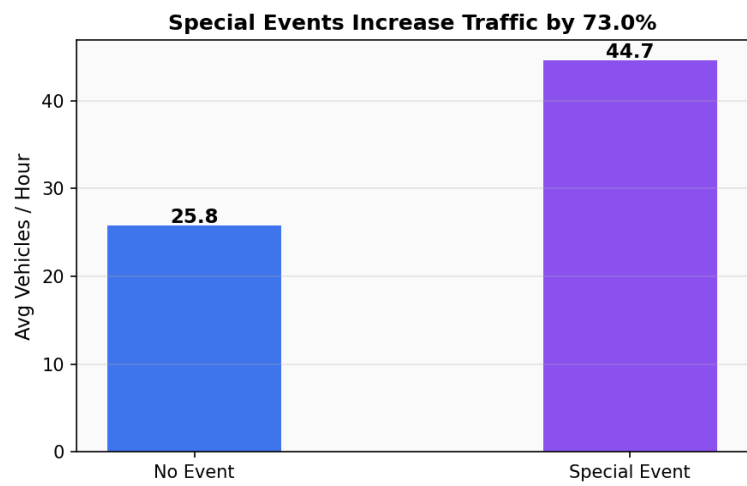
Weather Factor	Correlation with Vehicles	Average Volume	Business Impact
Rainfall (Heavy)	Positive (+)	32.6	+20.0% vs dry conditions

Temperature	Positive (+0.01)	27.7	Moderate effect
Humidity	Positive (+0.01)	27.7	Indirect (linked to rain)
Wind Speed	Positive (+0.00)	27.7	Minor effect

5. Special Events & Holiday Traffic Impact

Special events — including festivals, sports matches, concerts, and public holidays — create abnormal and often unpredictable traffic spikes. These spikes are localized to specific junctions and time windows, requiring targeted management responses.

Figure 6 — Special Event Impact on Average Traffic Volume



- Special events increase average vehicle count by 73.0% across all junctions.
- Junctions 1 (Central) and 3 (North) experience the highest event-related spikes, suggesting proximity to event venues.
- Holiday traffic patterns resemble weekend patterns but with elevated late-afternoon volumes.
- Events combined with rainfall produce the highest observed traffic volumes (Event_Impact: High).
- Post-event dispersal (30–90 minutes after event end) creates secondary congestion spikes.

6. Actionable Recommendations for Traffic Management

Based on the analysis of peak hours, patterns, and influencing factors, the following recommendations are proposed to mitigate congestion and optimize traffic flow across all junctions.

6.1 Peak Hour Management

Time-Based Signal Optimization	Implement adaptive traffic signal systems at Junctions 1 and 3 that automatically extend green phases during peak hours.
Staggered Access Lanes	Designate dedicated turning lanes and bus priority corridors at Junction 3 (North) during morning peak hours.
Demand Forecasting Alerts	Deploy predictive congestion alerts 20–30 minutes before expected peak onset using ML models (XGBoost).
Flexi-Hour Incentives	Collaborate with employers near Junctions 1 and 3 to promote staggered working hours — shifting 15% of staff to off-peak hours.

6.2 Weather-Responsive Management

Rain Activation Protocol	Pre-position additional Uber/cab resources at all four junctions when rainfall probability exceeds 40%.
Dynamic Signal Timing	Reduce signal cycle speeds by 10–15% during heavy rain to match the lower vehicle speeds and red
Weather-Triggered Alerts	Send push notifications to users near high-congestion junctions when rain is detected, recommending

6.3 Weekend & Late-Night Traffic

Entertainment Zone Management	Deploy traffic wardens at Junction 3 (North) on Friday and Saturday evenings (9 PM–1 AM) to manag
Night-Time Routing	Activate alternate route signage for late-night weekend traffic to distribute volume away from Junction
Weekend Bus Augmentation	Increase public transit frequency near Junction 2 (East) on Saturday/Sunday afternoons to absorb leis

6.4 Event & Holiday Management

Pre-Event Traffic Plans	For known events (concerts, festivals, matches), activate a junction-specific traffic management plan
Post-Event Dispersal Control	Designate one-way exit flows and temporary no-parking zones within 500m of event venues to accele
Event-Based Surge Coordination	Events increase traffic by 73% — notify ride-sharing drivers 1 hour before event end to pre-position ne
Holiday Calendar Integration	Integrate public holiday and major event calendars into the traffic signal management system to autom

7. Conclusion

This analysis confirms that traffic congestion at all four junctions follows predictable patterns driven by time-of-day, day-of-week, weather, and event factors. Junction 3 (North) represents the highest priority for intervention, followed by Junction 1 (Central).

- Peak hours (7–9 AM and 5–8 PM) increase traffic by 138% — targeted signal optimization during these windows will yield the highest congestion reduction.
- Rainfall events drive a 20% increase in vehicle volume — weather-responsive protocols and pre-deployed ride-sharing resources are the most cost-effective countermeasure.
- Special events cause 73% traffic spikes — predictable through calendar integration and manageable with pre-planned dispersal protocols.
- Weekend late-night traffic at Junctions 1 and 3 requires dedicated management distinct from weekday strategies — leisure patterns differ fundamentally from commuter patterns.
- Predictive ML models (LSTM or XGBoost) trained on this dataset can provide 20–30 minute advance congestion warnings, enabling proactive rather than reactive traffic management.

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